

shares and agent j dislikes. Via the same reasoning as in the previous paragraph, this is a contradiction, so $b_j^* < b_i^{**} < b_j^{**}$. Let us consider some prior $\hat{b} \in (b_i^{**}, b_j^{**})$, where agent i shares and agent j ignores. By symmetry of agent i and j 's network positions, it is clear that for agent i and agent j with prior $\hat{b}$ that $U_j(\mathcal{S}) - U_i(\mathcal{S}) = p_s \kappa$. Similarly, $U_j(\mathcal{I}) - U_i(\mathcal{I}) = 0$. Then notice that:

$$[U_j(\mathcal{S}) - U_i(\mathcal{S})] - [U_j(\mathcal{I}) - U_i(\mathcal{I})] = [U_j(\mathcal{S}) - U_j(\mathcal{I})] + [U_i(\mathcal{I}) - U_i(\mathcal{S})] = p_s \kappa > 0.$$

This implies that either $[U_j(\mathcal{S}) - U_j(\mathcal{I})] > 0$ or $[U_i(\mathcal{I}) - U_i(\mathcal{S})] > 0$ (or both). However, this is a contradiction because at prior b' , it is supposed to be a best response for agent j to play $\mathcal{I}$ and a best response for agent i to play $\mathcal{S}$. Thus, $b_i^{**} = b_j^{**}$. ■

Proof of Theorem 2. For part (a), let us consider belief $b^{(2)} < 1$ and a reliability threshold $\underline{r}$ such that for all $\mathbf{P}$, all agents with $b < b^{(2)}$ choose $\mathcal{D}$ in every equilibrium (including the most-sharing equilibrium) whenever the article has reliability $r < \underline{r}$. Such an $\underline{r}$ exists by Lemma A.4(ii). Thus, for all $r < \underline{r}$, every agent on an island $\ell \geq 2$ dislikes in the most-sharing equilibrium, regardless of $\mathbf{P}$.

Next, we consider an increase in homophily (while holding expected degree fixed). By our choice of $\underline{r}$, all agents on islands $\ell \geq 2$ still dislike in the most-sharing equilibrium whenever $r < \underline{r}$. We can thus consider the social media game that only involves island 1, treating islands 2 through k as automata that always dislike. Before the shift in homophily, consider the equilibrium cutoffs (b_1^*, b_1^{**}) for island 1 in the most-sharing equilibrium (the same for all agents on island 1, per Lemma 1) and let $\mathbf{B}_1$ denote the modified cutoff space defined by all cutoffs $(\hat{b}_1^*, \hat{b}_1^{**}) \preceq (b_1^*, b_1^{**})$. Finally we define a map $\varphi : \mathbf{B}_1 \rightarrow \mathbf{B}_1$ that maps cutoffs in $\mathbf{B}_1$, $(\hat{b}_1^*, \hat{b}_1^{**})$, to best-response cutoffs $(\hat{b}_1^{*,BR}, \hat{b}_1^{**,BR})$, given that agents on island 1 play according $(\hat{b}_1^*, \hat{b}_1^{**})$. By the arguments in Lemma A.2, φ preserves $\succeq$ and $\mathbf{B}_1$ is a complete sublattice, provided that the map φ is well-defined in that it always maps to an element in $\mathbf{B}_1$.

To establish this, consider the utility $U_1(\mathcal{S})$ of sharing on island 1 with homophily parameters (p_s, p_d) , holding fixed the cutoff strategy $(\hat{b}_1^*, \hat{b}_1^{**})$ and the expected degree of each agent on island 1, ζ_1 . Thus, we can write $p_d = (\zeta - N_1 p_s) / (N - N_1)$ and observe then that

$$U_1(\mathcal{S}) = U_1^{(1)} + \kappa N_1 p_s (1 - H(\hat{b}_1^{**})) - d \left(N_1 p_s H(\hat{b}_1^*) + \frac{\zeta - N_1 p_s}{N - N_1} \cdot (N - N_1) \right),$$

and in particular, $\partial U_1(\mathcal{S}) / \partial p_s = \kappa N_1 (1 - H(\hat{b}_1^{**})) + d N_1 (1 - H(\hat{b}_1^*)) > 0$. Therefore, if we compare utility $U_1'(\mathcal{S})$ after the increase in homophily to $U_1(\mathcal{S})$ before the increase in homophily (leaving $(\hat{b}_1^*, \hat{b}_1^{**})$ fixed), we see that $U_1'(\mathcal{S}) \geq U_1(\mathcal{S})$. Hence, φ necessarily maps any cutoffs in $\mathbf{B}_1$ into $\mathbf{B}_1$. Applying the same Tarski fixed-point theorem as before (see Tarski (1955)), the set of fixed points (and thus Bayesian-Nash equilibria) form a lattice within the space of cutoffs $\mathbf{B}_1$. Moreover, there is a most-sharing equilibrium in $\mathbf{B}_1$, which is also the most-sharing equilibrium in $\mathbf{B}$. We denote this equilibrium by $(b_1^{*'}, b_1^{**'})$ and note that $(b_1^{*'}, b_1^{**'}) \preceq (b_1^*, b_1^{**})$ (because it lies in $\mathbf{B}_1$). In particular, this means $b_1^{**'} \leq b_1^{**}$, and more agents share on island 1 in the most-sharing equilibrium following the rise in homophily.

To measure the change in virality, we first observe that the seed agent i^* (that maximizes $\mathbb{E}[\mathbf{S}_{i^*}]$)